Neural Representations of Face Recognition in Biological and Artificial Systems: Insights from MEG and CNNs

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Across time similarities on 3D brain maps.

Introduction

- Artificial Neural Networks performance have gone on to surpass human performance on many tasks.
- There is an increasing recognition of the added value of combining AI and neuroscience research to mutually reinforce one another.
- An emerging stream of research focuses on **comparing** artificial and biological systems' internal representations, across an array of cognitive tasks such as visual categorisation, as a framework to (a) **reverse engineer** the brain, (b) seek to build **biologically plausible** models[8,10,16,19,22].
- Faces represent a special type of object, and the brain is endowed with an efficient face recognition system that recruits multiple areas.
- This is tackled in some studies which in non-human primate mainly focus on the IT cortex, and in humans, largely rely on fMRI or behavioral data[6,7,12].

Goal

Compare the similarity of the internal representations of faces between the human brain and Convolutional Neural Networks:

- Neuromagnetic data (MEG) collected during face recognition
- Different CNN architectures
- Representational Similarity Analysis (RSA)

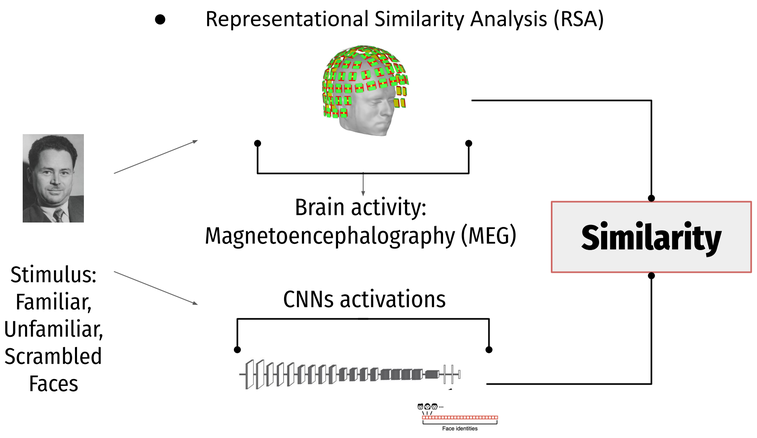

Figure 1. Study Overview

Methods: MEG data

- Used the: "multi-subject, multi-modal human neuroimaging dataset"[21].
- MEG data acquired using an Elekta Vectorview 306 system, N=16 subjects.
- **450 Facial stimuli**: unique identities (**150 Familiar, 150 Unfamiliar, 150 scrambled**), each presented twice: 900 trials.
- Preprocessing: Same as those in a previous study[5] with the same data (incl. filtering, rejection of bad trials).
- Selected **1 trial** per condition**: 428 trial** per subject (bad trials rejection).
- Epoched the data into segments of **1s**: 200ms before and 800ms after onset.
- Projected data to source space through computing:
  - BEM model.
  - The forward solution.
  - The inverse problem
  - Source estimates using "MNE" operator.

=> **1000ms** per voxel for **8092 voxel**.

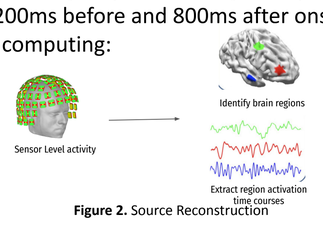

Figure 2. Source Reconstruction

Methods: Networks Training and activations extraction

- We selected 7 CNNs: **FaceNet**[17], **SphereFace**[14], **ResNet50**[4], **MobileNet**, **VGG16**[18] **Inception**[20], and **CORnet-S**[11].
- Trained each of the 7 models to solve 3 classification tasks:
  - **Face Recognition**: Trained on VGGFace[13] and fine-tuned on a Distribution of the the CelebA dataset (includes the 132 of the Familiar stimuli used in MEG).
  - **Object Recognition:** Trained on ImageNet.
  - **Dual-Task, Face and Object Recognition:** ImageNet with a face category.
- Extracting the activations of the networks:
  - Pass the facial Stimuli used in the MEG experiment (428 Stimuli).
  - Get the response of each layer for the 7 networks for the three classif. tasks.
  - Get the response of each layer for the 7 networks using **random weights**.

Methods: Representational Similarity Analysis (RSA)

- We used Representational Similarity Analysis[9] to assess the **similarity** between the activity patterns of the artificial and biological networks. We computed the Representational Dissimilarity Matrices (RDMs) as a measure of pairwise dissimilarities between the activations in response to the stimuli.

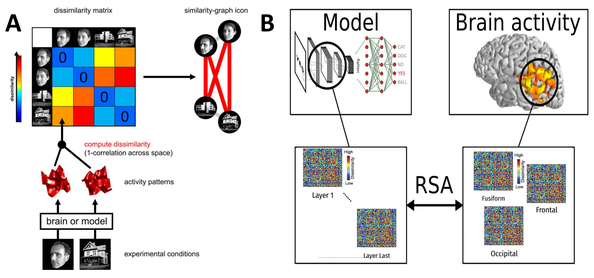

Figure 3. (A) Computing RDMs. **(B.)** Computing similarity values between MEG source reconstructed data and the models activations

- We computed **3 types of RDMs**: one for each type of stimuli (Familiar, Unfamiliar and Scrambled) for MEG and CNNs.
- **For networks**: 3 RDMs per layer for the 3 different tasks plus the untrained.
- **For brain data**:
  - Analysis I:
    - 3 RDMs per Region of Interest by taking the average signal of the voxels constituting each, **450 ROI** in total. RDMs are computed signal from **100ms** to **600ms**.
  - Analysis II:
    - 3 RDMs per time point for every ROI by taking the activity of all the voxels that constitute each ROI, **1000ms in total**.
- We computed the similarity between (i) RDMs of each layer for the networks trained on the 3 tasks as well as the untrained networks and: (ii) their respective RDMs in each ROI for analysis I **or** (iii) their respective RDMs for each time point of every ROI for analysis II
- Correlations in the RDM cells but also across RDMs (similarity scores) were calculated using pearson correlation.

Analysis I Results: MEG-ROIs similarities

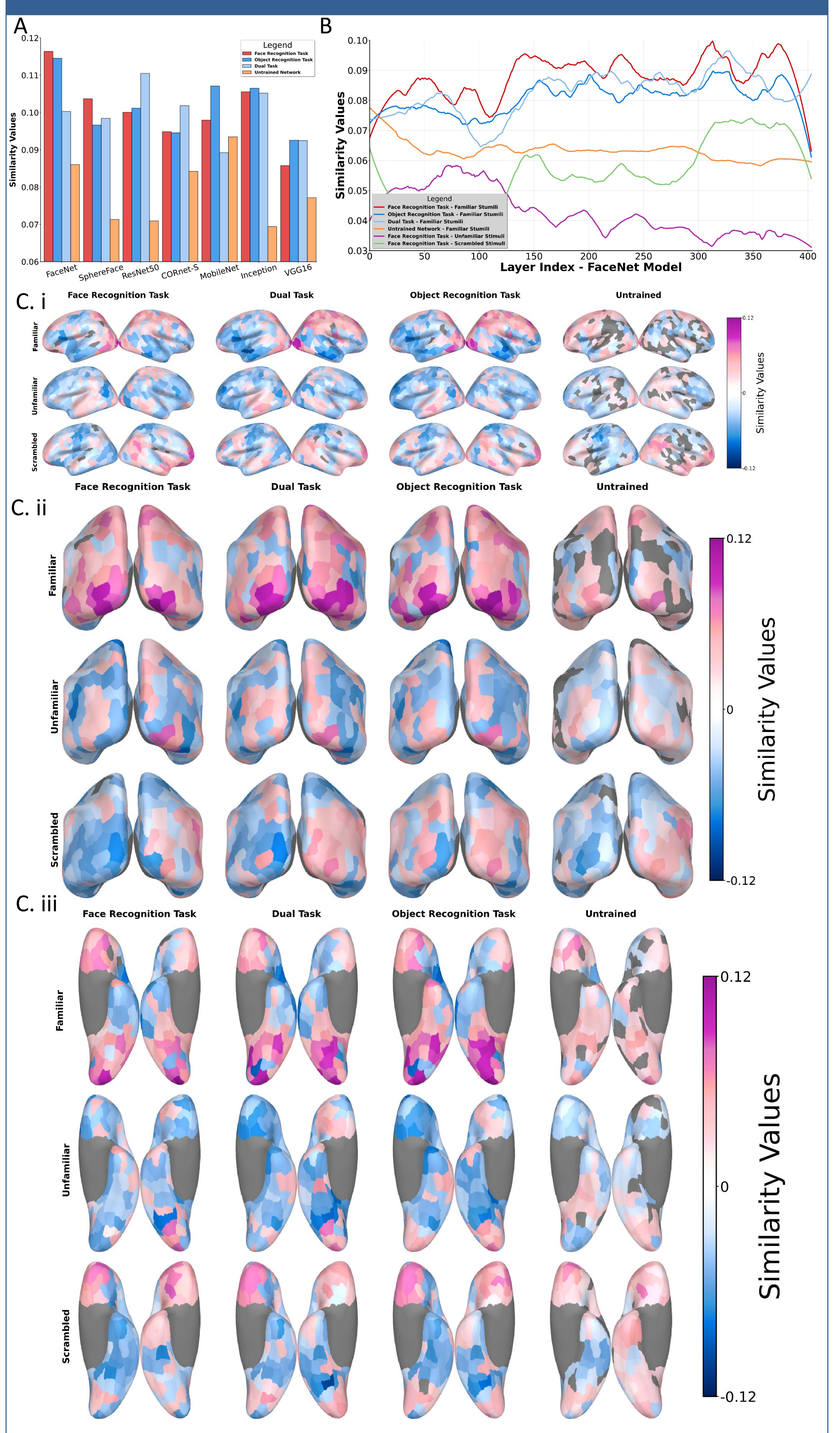

Figure 4.
(A) Maximum similarity scores obtained across all brain ROIs for each architecture's activations trained on the three different tasks, as well as the untrained architecture, when using Familiar RDMs (Familiar stimuli presented to models and subjects) ($p < 0.05$). **(B)** Maximum correlations between FaceNet layers trained on the three different tasks, as well as the untrained architecture, when using Familiar RDMs and the brain ROIs. Also, the maximum correlations between FaceNet layers trained on Face Recognition when using Scrambled and Unfamiliar RDMs and the brain ROIs ($p < 0.05$). **(C)** 3D brain maps of the MEG-FaceNet similarity scores by taking the maximum correlation between an ROI and FaceNet layers trained on the three classification tasks, in addition to the untrained FaceNet architecture, using the three types of RDMs ($p < 0.05$). **(i)** Caudal View. **(ii)** Lateral View. **(iii)** Ventral View.

- Training on any of the three tasks **improved** the similarity scores considerably compared to random weights: untrained architectures.
- For FaceNet and SphereFace, similarity scores obtained by the Face Recognition Task were higher than those obtained by the other tasks. For CORnet-S and ResNet50, similarity scores obtained by the Dual Task were the highest.
- **FaceNet exhibits the highest similarity scores**.
- Correlations levels for the trained FaceNet obtained using Familiar Input (Both participants and models know the ids) are higher than the three other experiments for the three classification tasks, indicating **higher similarities when recognizing Familiar Faces**.
- Similarity between the MEG activity and the last few layers drops at the end.
- 3D maps of the MEG-FaceNet similarity scores suggest **similarity peaks over occipital and, to some extent, intermediate areas,** specifically in Face Recognition Task using Familiar input.

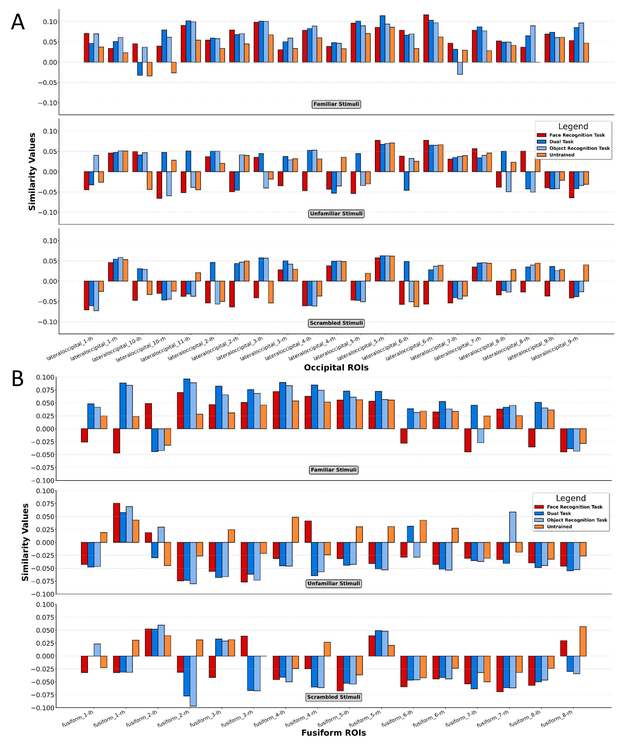

Figure 5. Extremum similarity scores obtained for specific ROIs for each FaceNet activations trained on the three different tasks, as well as the untrained architecture, when using Familiar the three types of RDMs ($p < 0.05$). **(A)** Fusiform ROIs **(B)** Occipital ROIs.

- In general, the Fusiform and Occipital areas exhibit notably high positive correlations when processing Familiar Stimuli. In contrast to the negative correlations observed with Unfamiliar and Scrambled Stimuli.
- **The exact face stimuli that are very similar in the ANN's eyes, are seen as very different by humans, and vice versa.**
- Fusiform regions 1-rh and 2-lf display distinct behaviors, indicating a potential stronger association with processing unfamiliar faces.

Analysis II Results: Across Time Similarities

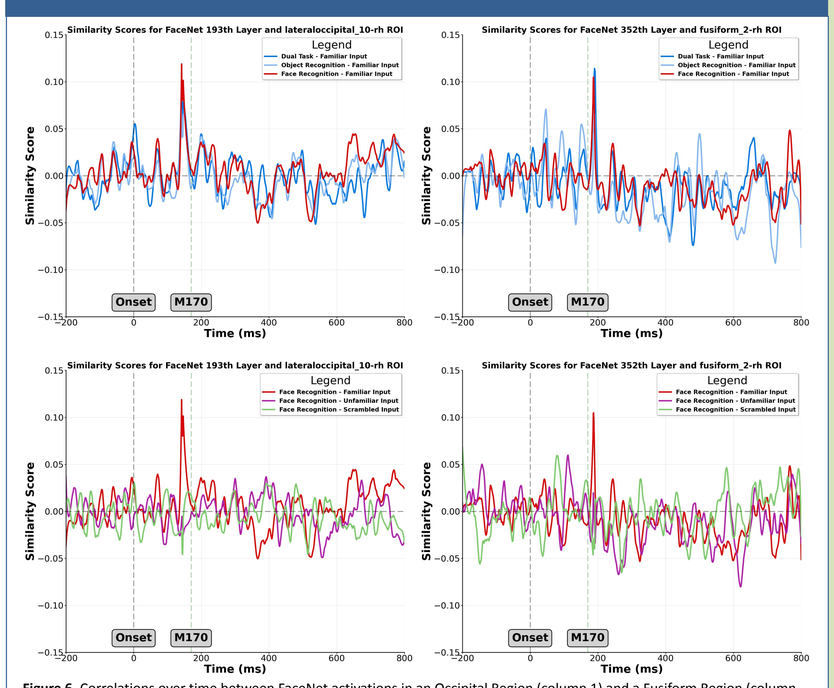

Figure 6. Correlations over time between FaceNet activations in an Occipital Region (column 1) and a Fusiform Region (column 2), showcasing variations across three distinct classification tasks (row 1) when exposed to Familiar Stimuli, as well as the divergences observed among the three stimulus types (row 2), all utilizing the FaceNet model trained specifically for Face Recognition.

- Negative and low similarity scores for the majority of timepoints.
- For the three classification tasks we have a peak of similarities when using the Familiar Faces RDMs.
- For the the Scrambled and Unfamiliar RDMs, no peak was observed.
- The occipital ROI peak took place around 5ms before the 170ms which coincides with the Event Related Potential for the task of face perception in the brain: M170.
- For the Fusiform ROI we notice an increase in similarity score that starts from 0 at 170 ms (M170) and reaches a peak >.12 at 180ms ($p<0.05$).

Conclusions

- ❑ Artificial Neural Networks (ANNs) display activations that appear to mirror the neuromagnetic patterns associated with the perception/recognition of Familiar Faces when trained specifically on the task of Face Recognition.
- ❑ Training focused solely on face recognition, as opposed to object recognition or a combination of both, proved essential and effective in achieving a notably high level of similarity between the model and brain patterns.
- ❑ When ANNs are fed Unfamiliar and Scrambled Faces as inputs, the resulting similarity to brain activity matches that of feeding Familiar Faces to untrained ANNs, indicating a specialization in Face Detection.
- ❑ Three-dimensional brain maps indicate prominent similarity peaks in the occipital areas and, to some extent, intermediate regions.
- ❑ Comparative analysis reveals a significant increase in similarity to FaceNet, trained specifically for face recognition, compared to an untrained model. This increase is particularly pronounced in certain regions of the Fusiform Face Area (FFA), rising from 0.02 to 0.08.
- ❑ Examining correlations over time reveals a notable alignment between ERP (Event-Related Potential) responses to face perception and peaks in similarity scores. These peaks are observed prominently in occipital and fusiform regions across the three classification tasks, with a slight advantage noted for the Face Recognition Task.
- ❑ In contrast, Scrambled and Unfamiliar Stimuli yield lower similarity scores in the analysis of correlations over time.

Future work

- ❑ Compute frequency bands and conduct the same two analyses across different bands.
- ❑ Explore the influence of loss functions on the model's similarity to brain patterns.
- ❑ Utilize visualization techniques to illustrate the specific aspects of stimuli that are most focused on by layers resembling brain activity.
- ❑ Evaluate alternative types of ANNs (GANs, Transformers, etc.)
- ❑ Examine the robustness of our findings, taking careful consideration of potential caveats and limitations inherent in the RSA approach[3].

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